Confidence Calibration under Ambiguous Ground Truth

Anonymous Author(s)
Affiliation
Address
email

Abstract

Confidence calibration implicitly assumes a unique ground-truth label per input. When multiple annotators genuinely disagree—as in medical diagnosis, natural language inference, and perceptual tasks—this assumption breaks down. We show that post-hoc calibrators fitted on majority-voted labels exhibit a systematic *calibration gap*: they appear well-calibrated against voted labels yet remain substantially miscalibrated against labels drawn from the annotator distribution. The gap is structural, affecting Temperature Scaling, Platt scaling, and Histogram Binning alike, because all share the same one-hot target. We prove that Temperature Scaling is biased toward lower temperatures than the annotator distribution requires, and that the gap grows monotonically with annotation entropy. To close it, we propose **Soft-Label Temperature Scaling (SLTS)** and related methods that optimize proper scoring rules against the full annotator distribution. Experiments on CIFAR-10H, ChaosNLI, ISIC 2019, and DermaMNIST—spanning seven architectures across vision and language—show that our methods reduce true-label ECE by 48–91% relative to standard Temperature Scaling.

1 Introduction

A well-calibrated classifier $f : \mathcal{X} \to \Delta^K$ produces probability vectors $\hat{p}(x) = f(x)$ such that stated confidence matches empirical accuracy: when $\hat{p}_{\hat{c}}(x) = p$ for the predicted class $\hat{c} = \arg\max_k \hat{p}_k(x)$, the true probability of $\hat{c}$ being correct is exactly p. Formally,

$$\mathbb{P}\big(\hat{Y} = Y \mid \hat{p}_{\hat{Y}}(X) = p\big) = p, \quad \forall\, p \in [0, 1], \tag{1}$$

where Δ^K denotes the K-dimensional probability simplex. Calibration is critical for safe deployment in medicine [Kompa et al., 2021], autonomous driving, and other high-stakes settings.

The hidden assumption. Equation (1) implicitly assumes a unique ground-truth label Y for every input x. In practice this assumption frequently fails: a skin lesion may be legitimately classified as malignant or benign by different dermatologists; a natural-language premise may or may not entail a hypothesis depending on pragmatic context; low-resolution objects can be categorically ambiguous. In such settings the correct target is not a single label but a *distribution over labels*, $\pi(\cdot \mid x) \in \Delta^K$, reflecting irreducible aleatoric uncertainty shared by rational annotators.

The standard practice and its failure. Standard practice aggregates annotations by majority vote to obtain a *voted label* y^* and calibrates against it. We show this produces a systematic *calibration gap*: the calibrator achieves low ECE against voted labels ($\mathrm{ECE}_{\text{voted}}$) while remaining substantially miscalibrated against true labels drawn from the annotator distribution ($\mathrm{ECE}_{\text{true}}$, following Stutz et al. [2023]). The mechanism is straightforward: ambiguous examples appear underconfident relative to the one-hot voted target, so Temperature Scaling selects a lower T than the annotator distribution requires, amplifying overconfidence. This failure is universal—Temperature Scaling, Platt scaling, and Histogram Binning all *widen* the gap in a controlled experiment (Section 4)—because the root cause is the voted-label target, not method capacity.

Contributions.

1. **Formal framework** (Section 3): we define true-label calibration and the calibration gap Δ with respect to the annotator distribution $\pi(\cdot \mid x)$.
2. **Motivating example** (Section 4): a controlled 3-class experiment in which all standard calibrators *widen* the gap, from $+7.6$ to $+9.5$–$+11.1$ pp.
3. **Theory** (Section 5): we prove that TS is biased toward lower temperatures than the annotator distribution requires, and that the gap grows monotonically with annotation entropy.
4. **Methods** (Section 6): post-hoc calibrators—SLTS, MCTS, VS, IR-Soft, SoftPlatt, and Dirichlet-Soft—that optimize proper scoring rules against annotator distributions; we further show (Appendix F) that the improvement is due to the soft target, not additional model capacity.
5. **Experiments** (Section 7): evaluation on four benchmarks (CIFAR-10H, ChaosNLI, ISIC 2019, DermaMNIST in appendix) across seven architectures spanning vision and language, confirming the gap is modality-agnostic.

2 Related Work

Confidence calibration. Guo et al. [2017] showed that modern networks are miscalibrated and that Temperature Scaling (TS) is an effective post-hoc remedy. Isotonic regression [Zadrozny and Elkan, 2002], Platt scaling [Platt, 1999], and Dirichlet calibration [Kull et al., 2019] have since been studied extensively—all assuming a unique ground-truth label. Minderer et al. [2021] further showed that Vision Transformers are better calibrated than CNNs, but the same voted-label assumption underlies all evaluations. Recent work by Bai et al. [2021] and Tian and Xiao [2023] revisits the sufficiency of Temperature Scaling, but continues to calibrate against single-label supervision. We show the shared assumption—not method capacity—is the limiting factor.

Learning from multiple annotators. Methods for inferring labels from crowds [Dawid and Skene, 1979, Rodrigues and Pereira, 2018], datasets capturing annotator disagreement [Peterson et al., 2019, Nie et al., 2020], and surveys of multi-annotator learning [Uma et al., 2021, Collins et al., 2022, Aroyo and Welty, 2015] have received growing attention. Gordon et al. [2022] and Plank [2022] highlight the epistemological problems with annotation aggregation in NLP. None of these works study the interaction between annotator disagreement and post-hoc calibration.

Conformal prediction under ambiguous ground truth. Stutz et al. [2023] show that conformal sets calibrated on voted labels undercover the true annotator distribution. We identify the analogous failure for confidence calibration: voted-label calibration produces systematic overconfidence. Angelopoulos et al. [2024] provides a broader survey of conformal methods; our work extends this research program to the calibration setting.

Calibration under distribution shift and soft labels. Calibration under covariate shift [Park et al., 2020, Wang et al., 2020, Mukhoti et al., 2020] is orthogonal to our label-ambiguity setting. Label smoothing [Szegedy et al., 2016, Müller et al., 2019] applies uniform soft targets during training; we use annotation-informed soft labels for post-hoc calibration. Zhang et al. [2021] and Thulasidasan et al. [2019] study training-time soft label methods, while we focus on post-hoc calibration which requires no retraining.

3 Problem Formulation

3.1 Setup and Notation

Let $\mathcal{X}$ be an input space and $\mathcal{Y} = \{1, \dots, K\}$ a label space with K classes. A classifier $f : \mathcal{X} \to \Delta^K$ outputs a probability vector $\hat{p}(x) = f(x)$, typically computed as $\mathrm{softmax}(z(x))$ where $z(x) \in \mathbb{R}^K$

are pre-softmax logits. For each input x, the **annotator label distribution** $\pi(\cdot \mid x) \in \Delta^K$ gives the probability that a randomly drawn rational annotator assigns each label. Given m independent annotations $\{a_1, \dots, a_m\} \subset \mathcal{Y}$ for input x, we estimate

$$\hat{\pi}_k(x) = \frac{1}{m} \sum_{j=1}^{m} \mathbf{1}[a_j = k], \qquad k = 1, \dots, K. \tag{2}$$

The **voted label** is $y^* = \arg\max_k \hat{\pi}_k(x)$. On a calibration set $\{(x_i, z_i, \hat{\pi}(x_i))\}_{i=1}^n$ of n examples with logits $z_i \in \mathbb{R}^K$ and empirical soft labels $\hat{\pi}(x_i)$, the voted label is $y_i^* = \arg\max_k \hat{\pi}_k(x_i)$.

3.2 Voted-Label Calibration and Temperature Scaling

Standard calibration (Eq. 1) uses the voted label $Y = y^*$. Temperature Scaling [Guo et al., 2017] optimizes a single scalar $T > 0$ by minimizing cross-entropy on the calibration set:

$$\mathcal{L}_{\text{TS}}(T) = -\frac{1}{n} \sum_{i=1}^{n} \log \operatorname{softmax}(z_i/T)_{y_i^*}. \tag{3}$$

3.3 True-Label Calibration

Following Stutz et al. [2023], we distinguish between the **voted label** $y^* = \arg\max_k \hat{\pi}_k(x)$ and the **true label** $\tilde{y} \sim \pi(\cdot \mid x)$, a label drawn from the annotator distribution. Standard calibration (1) evaluates against voted labels; we define true-label calibration by evaluating against true labels.

Definition 1 (True-Label Calibration)**.** *A classifier f is **true-label calibrated** if, for the predicted class $\hat{c}(x) = \arg\max_k \hat{p}_k(x)$ and all confidence levels $p \in [0, 1]$:*

$$\mathbb{P}\big(\tilde{Y} = \hat{c}(X) \mid \hat{p}_{\hat{c}}(X) = p\big) = p, \quad \tilde{Y} \sim \pi(\cdot \mid X). \tag{4}$$

True-label calibration reduces to standard calibration (1) when $\pi(\cdot \mid x)$ is always one-hot.

We define two ECE metrics. The **voted-label ECE** $\text{ECE}_{\text{voted}}$ is the standard ECE computed against the voted label y^*. The **true-label ECE** ECE_{true} is computed by drawing a single label $\tilde{y}_i \sim \pi(\cdot \mid x_i)$ for each test example, computing the standard ECE against these true labels, and averaging over 100 draws to reduce variance. Both use $B = 15$ equal-width bins.

3.4 The Calibration Gap

Definition 2 (Calibration Gap)**.** *The **calibration gap** of a classifier f on a dataset with annotator distribution $\hat{\pi}$ is*

$$\Delta = \text{ECE}_{\textit{true}} - \text{ECE}_{\textit{voted}}, \tag{5}$$

where $\text{ECE}_{\textit{voted}}$ evaluates against the voted label y^* and $\text{ECE}_{\textit{true}}$ evaluates against true labels $\tilde{y} \sim \hat{\pi}(\cdot \mid x)$.

A positive Δ indicates that the model is more miscalibrated under true-label evaluation than under voted-label evaluation—it appears well-calibrated on voted labels while being overconfident with respect to the annotator distribution.

4 Motivating Example

Setup. A 3-class 2D Gaussian dataset: class 0 is always labeled 0 ($\pi_0 = 1$); class 1 has annotator distribution $\pi(x) = [0, 0.70, 0.30]$ (voted label always 1); class 2 is always labeled 2 ($\pi_2 = 1$). A 2-layer MLP (128 hidden units) is trained on voted labels. This is the simplest setting that exposes the calibration gap: one cluster is perfectly unambiguous to the annotators but the voted label is always the majority class.

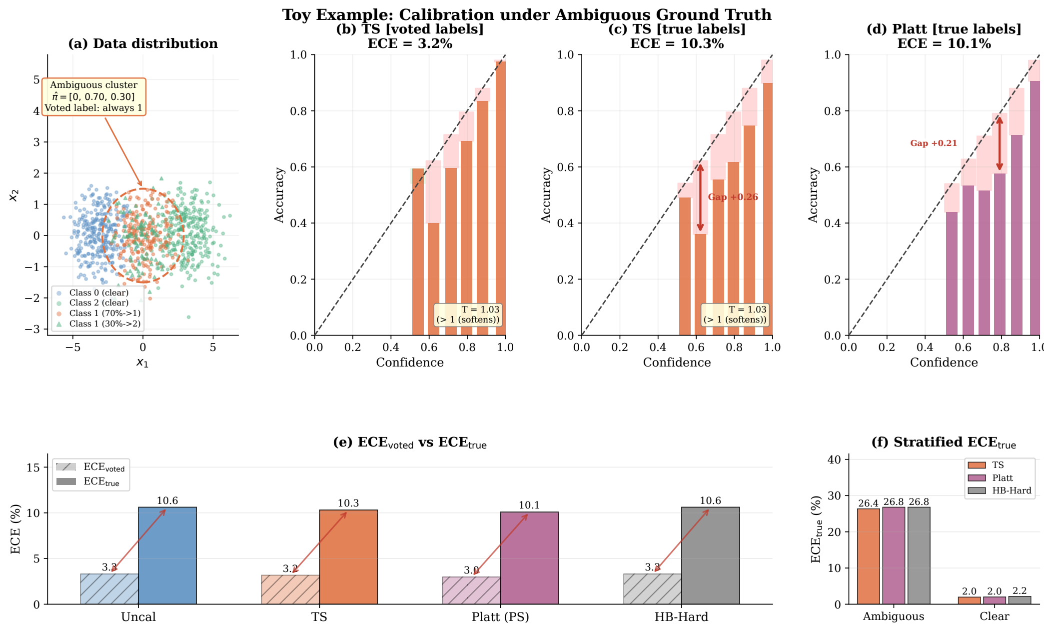

Figure 1: **Motivating example: all standard calibration methods fail.** **(a)** Data distribution: the middle cluster is ambiguous ($\hat{\pi} = [0, 0.70, 0.30]$; voted label always Class 1). **(b)** TS against voted labels: appears well-calibrated ($\text{ECE}_{\text{voted}} = 3.2\%$). **(c)** TS against true labels: large calibration gap ($\text{ECE}_{\text{true}} = 10.3\%$, $\Delta = +7.1$ pp). **(d)** Platt scaling against true labels: also fails ($\text{ECE}_{\text{true}} = 10.1\%$), confirming the gap is independent of method capacity. **(e)** All voted-label baselines show $\text{ECE}_{\text{true}} \approx 10$–$12\%$ and $\Delta \approx +8$–11 pp; reducing $\text{ECE}_{\text{voted}}$ does not reduce ECE_{true}. **(f)** Stratified ECE_{true}: the gap is concentrated in the ambiguous cluster and persists for all voted-label methods.

Results. Table 1 and Figure 1 show that the calibration gap is a structural consequence of the voted-label target. TS achieves $\text{ECE}_{\text{voted}} = 1.1\%$ but $\text{ECE}_{\text{true}} = 12.2\%$ ($\Delta = +11.1$ pp)—*larger* than the uncalibrated gap of $+7.6$ pp. TS sets $T = 0.76 < 1$, boosting confidence toward the voted target and thereby pushing the model *further* from the true annotator distribution. Platt scaling ($\Delta \approx +11$ pp) and even non-parametric Histogram Binning ($\Delta = +9.5$ pp) exhibit the same pathology, confirming that the failure lies in the target, not the method.

Remark 1 (Temperature direction in the toy example)**.** *TS finds $T = 0.76 < 1$ in the toy example, consistent with Proposition 1. The ambiguous cluster (class 1, $\pi = [0, 0.55, 0.45]$) has voted label always equal to class 1, so the model achieves near-perfect voted-label accuracy while outputting only moderate confidence ($\approx 75\%$); TS interprets this as underconfidence and sets $T < 1$ to boost it. A calibrator targeting the annotator distribution would instead need $T > 1$ to reduce confidence toward the true target of* 55%.

Table 1: **Motivating toy example: all voted-label calibration methods widen the calibration gap.** Every voted-label method reduces $\text{ECE}_{\text{voted}}$ but *increases* ECE_{true}, growing the gap $\Delta = \text{ECE}_{\text{true}} - \text{ECE}_{\text{voted}}$ from $+7.6$ pp (uncalibrated) to $+9.5$–$+11.1$ pp. TS sets $T = 0.76 < 1$ (wrong direction for ambiguous data). The failure is structural: it stems from the voted-label target, not method capacity.

Method	T	$\text{ECE}_{\text{voted}}$ (%) ↓	ECE_{true} (%) ↓	Gap Δ (%)
Uncalibrated	—	2.8	10.4	+7.6
TS	0.76	**1.1**	12.2	+11.1
Platt (PS)	—	1.4	12.2	+10.8
HB-Hard	—	2.5	12.1	+9.5

5 Theory: Why Standard Calibration Fails

We now formalize two complementary aspects of the calibration gap: the *direction* of the TS bias (Proposition 1) and its *scaling* with label ambiguity (Proposition 2).

Proposition 1 (Direction of TS Bias). *Consider a calibration set containing an ambiguous cluster in which the voted label y^* equals the majority class for every example, but the true annotator probability of y^* is $\pi_{y^*}(x) = q < 1$. If the model is already reasonably accurate on this cluster ($\mathrm{softmax}(z_i)_{y^*} > q$), then TS minimizing $\mathcal{L}_{TS}$ finds $T^* < 1$ (increasing model confidence), whereas soft calibration requires $T^* > 1$ (reducing confidence to q).*

Proof sketch. On the ambiguous cluster, all voted labels are y^*, so the voted-label loss (3) is minimized by pushing $\mathrm{softmax}(z_i/T)_{y^*} \to 1$, i.e. $T \to 0$. Hence $\partial\mathcal{L}_{\mathrm{TS}}/\partial T > 0$ and the optimal $T^* < 1$. Soft calibration instead minimizes $\mathcal{L}_{\mathrm{SLTS}}$ with target $q < 1$; the minimum requires $\mathrm{softmax}(z_i/T^*)_{y^*} \approx q$, necessitating $T^* > 1$ whenever $\mathrm{softmax}(z_i)_{y^*} > q$. Full proof in Appendix B. $\square$

Proposition 2 (Gap Monotonicity in Annotation Entropy). *Let $H(x) = -\sum_k \pi_k(x) \log \pi_k(x)$ denote the annotation entropy of x. Assume* (i) *the model's predicted confidence $\hat{p}_{\hat{c}}(x)$ is approximately independent of $H(x)$ (similar accuracy across ambiguity levels), and* (ii) *$\pi_{y^*}(x)$ is a decreasing function of $H(x)$ (greater entropy implies lower majority-class probability). Then the expected per-example contribution to Δ is non-decreasing in $H(x)$.*

Proof sketch. For unambiguous examples ($H(x) = 0$), $\pi(\cdot \mid x)$ is one-hot, so $\mathrm{ECE}_{\text{true}} = \mathrm{ECE}_{\text{voted}}$ and the contribution to Δ is zero. As $H(x)$ increases, $\pi_{y^*}(x)$ decreases below 1. The true-label calibration error for the predicted class is $|\hat{p}_{\hat{c}}(x) - \pi_{\hat{c}}(x)|$; since the model is trained against the voted label (targeting ≈ 1), $\hat{p}_{\hat{c}}(x) > \pi_{y^*}(x)$, and this overconfidence gap grows as $\pi_{y^*}(x)$ falls. The voted-label calibration error $|\hat{p}_{\hat{c}}(x) - \mathbf{1}[\hat{c} = y^*]|$ is unaffected by ambiguity because the voted label remains y^*. Hence Δ grows with $H(x)$. $\square$

6 Methods

All methods are *post-hoc*: they operate on cached logits from a fixed pre-trained model and require only a calibration set equipped with annotator distributions or individual annotations. No retraining is needed.

6.1 Soft-Label Temperature Scaling (SLTS)

SLTS minimizes the cross-entropy between the calibrated output and the soft label distribution:

$$\begin{aligned} \mathcal{L}_{\mathrm{SLTS}}(T) &= -\frac{1}{n} \sum_{i=1}^{n} \sum_{k=1}^{K} \hat{\pi}_k(x_i)\ \log \mathrm{softmax}(z_i/T)_k \\ &= \frac{1}{n} \sum_{i=1}^{n} \mathrm{KL}\big(\hat{\pi}(x_i) \,\|\, \mathrm{softmax}(z_i/T)\big) + \text{const.} \end{aligned} \tag{6}$$

SLTS uses the same single-parameter family as TS but with a target that correctly reflects the annotator distribution.

Proposition 3 (Proper scoring rule). *$\mathcal{L}_{SLTS}$ is a proper scoring rule with respect to $\hat{\pi}$: it is uniquely minimized (over all distributions, not just the one-parameter TS family) when $\mathrm{softmax}(z/T) = \hat{\pi}(x)$ for every x.*

6.2 Monte Carlo Temperature Scaling (MCTS)

MCTS is the natural method when annotator data arrives as individual annotation records rather than pre-aggregated distributions: each annotator's vote is treated as a sample from the latent annotator distribution, and the calibration loss is averaged over all such samples. For each calibration example x_i, we draw S labels $a_{i1}, \ldots, a_{iS} \sim \hat{\pi}(x_i)$ and minimize

$$\mathcal{L}_{\text{MCTS}}(T) = -\frac{1}{nS}\sum_{i=1}^{n}\sum_{s=1}^{S}\log\operatorname{softmax}(z_i/T)_{a_{is}}. \tag{7}$$

As $S \to \infty$, $\mathcal{L}_{\text{MCTS}} \to \mathcal{L}_{\text{SLTS}}$ in expectation, with variance $\propto 1/S$ by the law of large numbers. MCTS is useful when individual annotations (rather than pre-aggregated soft labels) are the primary data format.

6.3 Vector Scaling with Soft Labels (VS)

Vector Scaling extends TS by learning a per-class temperature $\mathbf{T} = (T_1, \ldots, T_K) \in \mathbb{R}^K_{>0}$, optimizing $\mathcal{L}_{\text{SLTS}}$ with $\operatorname{softmax}(z_k/T_k)$. Per-class temperatures accommodate datasets where some classes are systematically more ambiguous than others; this is particularly useful for the ISIC 2019 benchmark where the MEL/NV pair is far more ambiguous than DF/VL (confirmed by the per-class analysis in Figure 4).

6.4 Soft-Label Non-Parametric Methods

Soft Isotonic Regression (IR-Soft). Fits a monotone mapping from predicted confidence to mean annotator accuracy via the Pool Adjacent Violators Algorithm (PAVA), with soft targets. Non-parametric and fully automatic.

6.5 Additional Soft-Label Methods

Soft Platt Scaling (SoftPlatt). The natural soft-label extension of Platt / Matrix Scaling: the diagonal affine transform $W \cdot z + b$ (same structure as Platt, with per-class weight and bias) fitted against the annotator distribution via KL divergence. This method isolates the contribution of the soft target from any architectural difference: if SoftPlatt matches SLTS, it confirms that a single temperature is sufficient; if it outperforms, per-class scaling provides additional benefit.

Dirichlet-Soft. Extends Dirichlet calibration [Kull et al., 2019] to use annotator distributions as targets. The full $K \times K$ weight matrix W and bias b are fitted by minimizing $\text{KL}(\hat{\pi}(x) \| \operatorname{softmax}(Wz + b))$ with off-diagonal L_2 regularisation (ODIR, $\lambda = 10^{-3}$). This is the most expressive parametric soft-label calibrator; we include it to verify that the benefit of soft targets is not a capacity effect but a target effect (Appendix F).

7 Experiments

We evaluate on three main benchmarks with multi-annotator data, each with two backbones: CIFAR-10H (ResNet-50, ViT-B/16), ChaosNLI (RoBERTa-Large, DeBERTa-v3), and ISIC 2019 (EfficientNet-B4, ViT-S/16). DermaMNIST (ResNet-18, ViT-S/16) appears in Appendix H.

Baselines. We compare against (i) **Uncalibrated**: raw softmax; (ii) **TS**: Temperature Scaling on voted labels [Guo et al., 2017]; (iii) **Platt (PS)**: per-class weight and bias on logits, calibrated against voted labels [Platt, 1999, Kull et al., 2019]; (iv) **Dirichlet-Hard**: Dirichlet calibration [Kull et al., 2019] with the full $K \times K$ weight matrix and bias, calibrated against voted labels—the most expressive standard post-hoc calibrator; (v) **HB-Hard**: equal-width Histogram Binning [Zadrozny and Elkan, 2001] with voted labels.

Metrics. All metrics are evaluated against true labels $\tilde{y} \sim \pi(\cdot \mid x)$: ECE_{true} (averaged over 100 draws; $B = 15$ equal-width bins), adaptive ECE (aECE_t; equal-mass bins, same draws), classwise ECE (cwECE_t; per-class ECE averaged over K classes [Kull et al., 2019]), Brier score (Br_t), and NLL (NLL_t). After Table 2 we abbreviate these as ECE, aECE, cwECE, Br, and NLL.

Table 2: **Main results on CIFAR-10H.** $\mathrm{ECE}_{\text{true}}$: true-label ECE; Br_t/NLL_t: true-label Brier score/NLL. Soft-label methods achieve substantially better true-label calibration across both architectures. Best per architecture in **bold**.

Method	T	$\mathrm{ECE}_{\text{true}}$ (%) $\downarrow$	aECE_t (%) $\downarrow$	cwECE_t (%) $\downarrow$	$\mathrm{Br}_t \downarrow$	$\mathrm{NLL}_t \downarrow$
ResNet-50 — Voted-label baselines						
Uncalibrated	—	4.97	5.71	1.09	0.120	0.692
TS	2.03	4.29	4.25	0.91	0.112	0.363
Platt (PS)	—	4.29	4.23	0.91	0.112	0.372
Dirichlet-Hard	—	4.46	4.44	0.95	0.114	0.395
HB-Hard	—	7.44	7.34	7.97	0.516	10.40[†]
ResNet-50 — Soft-label methods (ours)						
MCTS	3.18	1.51	1.39	0.45	0.110	0.293
SLTS	3.18	1.51	1.39	0.45	0.110	0.293
SoftPlatt	—	1.52	1.31	**0.35**	0.110	0.288
VS	—	1.35	1.26	0.39	**0.109**	0.289
Dirichlet-Soft	—	1.25	1.06	**0.35**	**0.109**	**0.271**
IR-Soft	—	**0.72**	**0.83**	0.45	0.115	0.340
ViT-B/16 — Voted-label baselines						
Uncalibrated	—	4.99	5.36	1.06	0.111	0.678
TS	2.04	4.48	4.40	0.93	0.106	0.346
Platt (PS)	—	4.54	4.38	0.93	0.105	0.363
Dirichlet-Hard	—	4.70	4.62	0.97	0.107	0.394
HB-Hard	—	6.55	6.44	5.84	0.388	7.37[†]
ViT-B/16 — Soft-label methods (ours)						
MCTS	3.07	0.85	0.57	0.39	0.102	0.278
SLTS	3.07	0.85	0.56	0.39	0.102	0.278
SoftPlatt	—	0.88	0.47	**0.24**	0.101	0.272
VS	—	0.92	0.50	0.32	0.101	0.274
Dirichlet-Soft	—	**0.72**	**0.41**	0.25	**0.100**	**0.255**
IR-Soft	—	0.91	0.61	0.43	0.105	0.321

[†] HB assigns $(1-c)/(K-1)$ to non-predicted classes; large NLL/Br reflects this for $K=10$.

7.1 CIFAR-10H

Dataset and models. CIFAR-10H [Peterson et al., 2019] provides $\approx$51 human annotations per image for the 10 000-image CIFAR-10 test set. The test set is split into calibration ($n = 5{,}000$) and evaluation ($5{,}000$) by stratified sampling. We evaluate two architectures: **ResNet-50** [He et al., 2016] pretrained on ImageNet-1k, fine-tuned for 30 epochs (AdamW, cosine annealing), achieving 97.3% top-1 accuracy. **ViT-B/16** [Dosovitskiy et al., 2021] pretrained on ImageNet-21k, fine-tuned with the same protocol, achieving 98.1% top-1 accuracy.

Main results. Table 2 reveals a consistent pattern across both backbones. Voted-label baselines leave substantial residual ECE (TS: 4.29% for ResNet-50, 4.48% for ViT-B/16); crucially, Dirichlet-Hard—despite its $K \times K$ parameter matrix—does not improve over TS (4.46% and 4.70% respectively), confirming that the issue lies in the voted-label target, not the method's expressiveness. Soft-label methods close the gap decisively: SLTS reduces ECE to 1.51% and 0.85% respectively (65–81% relative reduction), and IR-Soft achieves the lowest ECE (0.72%) on ResNet-50. Dirichlet-Soft achieves the best ECE on ViT-B/16 (0.72%) and the best Brier score and NLL across both architectures, confirming that the expressive $K \times K$ parameterization is beneficial when combined with a soft target. Figure 2 plots $\mathrm{ECE}_{\text{voted}}$ vs. $\mathrm{ECE}_{\text{true}}$ for all methods, illustrating the gap between the two metrics.

Stratified analysis. Figure 3 shows ECE by annotation entropy quartile. For unambiguous inputs (Q1), all methods perform similarly (ECE $\approx$ 1–2%), since true-label calibration reduces to voted-label calibration when $\pi(\cdot \mid x)$ is one-hot. For highly ambiguous inputs (Q4), TS degrades to ECE $\approx 18\%$ while SLTS maintains $\approx 6\%$—a $3\times$ gap, directly confirming Proposition 2.

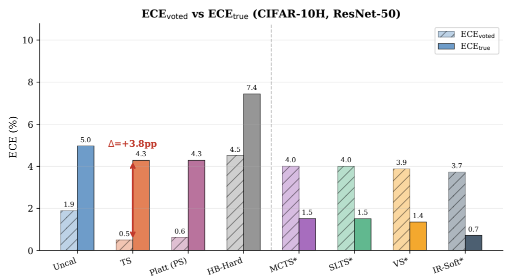

Figure 2: ECE$_{\textbf{voted}}$ **vs.** ECE$_{\textbf{true}}$ **(CIFAR-10H).** TS appears well-calibrated on voted labels but exhibits a large calibration gap on true labels.

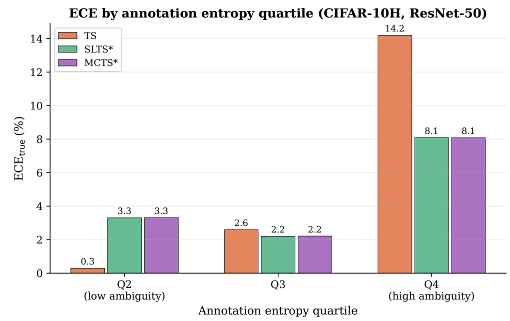

Figure 3: **ECE by annotation entropy quartile (CIFAR-10H).** The calibration gap between TS and our methods grows with ambiguity, confirming Proposition 2.

Table 3: **Main results on ChaosNLI** (combined SNLI+MNLI, 100 annotators per example). ECE/Br/NLL are all true-label metrics (see Table 2). Soft-label methods achieve substantially better calibration for both NLI architectures.

Method	T	ECE (%) ↓	aECE (%) ↓	cwECE (%) ↓	Br ↓	NLL ↓
RoBERTa-Large — Voted-label baselines						
Uncalibrated	—	27.79	27.77	18.61	0.650	1.384
TS	2.42	10.55	10.38	7.30	0.537	0.886
Platt (PS)	—	11.16	10.92	7.28	0.530	0.875
Dirichlet-Hard	—	11.55	11.40	7.64	0.535	0.889
HB-Hard	—	8.37	8.22	8.33	0.562	0.963
RoBERTa-Large — Soft-label methods (ours)						
MCTS	3.41	3.23	2.59	4.62	0.520	0.862
SLTS	3.41	3.22	2.59	4.62	0.520	0.862
SoftPlatt	—	2.66	2.32	2.43	0.509	0.839
VS	—	3.46	3.07	5.13	0.518	0.857
Dirichlet-Soft	—	**2.57**	**1.71**	**2.03**	**0.510**	**0.839**
IR-Soft	—	2.65	2.59	6.24	0.549	0.934
DeBERTa-v3-base — Voted-label baselines						
Uncalibrated	—	35.97	35.92	24.02	0.743	2.148
TS	3.81	11.63	11.57	8.48	0.549	0.900
Platt (PS)	—	12.43	12.36	8.43	0.539	0.896
Dirichlet-Hard	—	12.69	12.63	8.84	0.539	0.895
HB-Hard	—	10.75	10.26	9.71	0.574	0.989
DeBERTa-v3-base — Soft-label methods (ours)						
MCTS	5.27	3.49	3.38	6.56	0.529	0.876
SLTS	5.28	3.45	3.36	6.53	0.529	0.877
SoftPlatt	—	3.17	2.92	3.18	0.511	0.841
VS	—	3.91	3.35	6.46	0.525	0.869
Dirichlet-Soft	—	3.19	**2.71**	**2.95**	**0.509**	**0.836**
IR-Soft	—	**2.15**	3.96	7.20	0.554	0.942

7.2 ChaosNLI: Natural Language Inference

We next evaluate on **ChaosNLI** [Nie et al., 2020], an NLI benchmark with dense human annotations.

Dataset and models. ChaosNLI provides 100 human annotations per example for subsets of the SNLI [Bowman et al., 2015] and MultiNLI [Williams et al., 2018] dev sets. Each annotator assigns one of 3 labels (entailment, neutral, contradiction). We use the combined SNLI+MultiNLI subset ($n = 3{,}113$ examples), split 50/50 into calibration and test sets stratified by voted label. We evaluate two pre-trained NLI models: **RoBERTa-Large** [Liu et al., 2019] fine-tuned on MNLI. **DeBERTa-v3-base** [He et al., 2023] fine-tuned on MNLI.

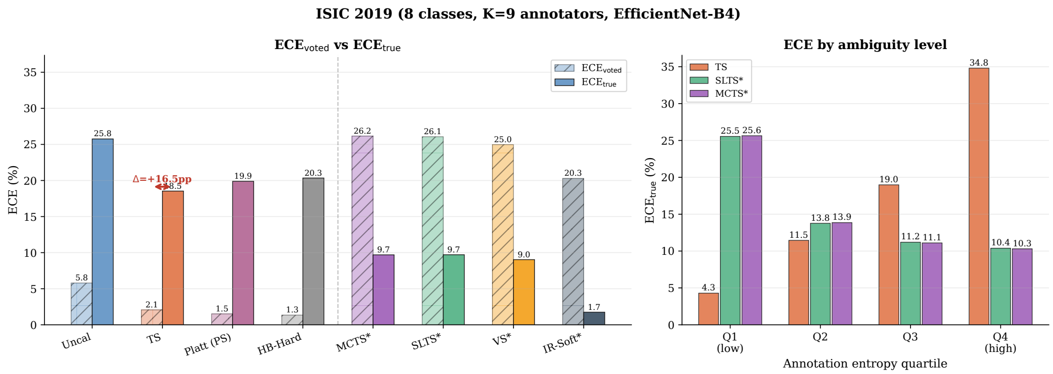

Figure 4: **ISIC 2019 results** (inspired by the Liu et al. [2020] reader study). **Left:** $\mathrm{ECE}_{\text{voted}}$ vs. ECE: voted-label baselines show a large calibration gap; soft-label methods reduce ECE substantially. **Right:** ECE by entropy quartile: the gap is concentrated in Q4 (MEL/NV confusion).

Results. Table 3 confirms the phenomenon extends to NLI. Dirichlet-Hard again fails to improve over TS (11.55% vs. 10.55% for RoBERTa; 12.69% vs. 11.63% for DeBERTa), reinforcing that the failure is in the target, not method capacity. SLTS reduces ECE to 3.22% and 3.45% (69–70% reduction), IR-Soft reaches 2.65% and 2.15%, and Dirichlet-Soft achieves the best Brier score and NLL for both architectures (0.510/0.509 and 0.839/0.836 respectively), demonstrating that expressive soft-label calibrators provide additional gains over the temperature-only baseline. The temperature direction is consistent with Proposition 1: SLTS requires substantially higher T than TS (3.41 vs. 2.42 for RoBERTa; 5.28 vs. 3.81 for DeBERTa), confirming the bias extends to language models.

7.3 ISIC 2019: Skin Disease Diagnosis

Inspired by the reader study of Liu et al. [2020] (9 dermatologists, 73% inter-reader agreement), we simulate $K = 9$ annotations per image via a clinically calibrated confusion matrix (Appendix I) on the **ISIC 2019** dataset [Codella et al., 2019, Combalia et al., 2019] (8 skin conditions, ≈25,000 images).

Annotator model. We simulate $K = 9$ dermatologist annotations per image using a clinically calibrated 8×8 confusion matrix (Appendix I). Agreement rates match Liu et al. [2020] and Haenssle et al. [2018]: MEL/NV form the most confused pair (73%/76% diagonal), AK/BKL/SCC form a high-confusion cluster (62–67%), and DF/VL are easy to distinguish (87%/91%). Weighted overall agreement is 73%, matching Liu et al. [2020] exactly.

Setup. We evaluate two architectures on a stratified 70%/15%/15% train/val/test split (≈3,750 calibration and test images each) with class-weighted cross-entropy. **EfficientNet-B4** [Tan and Le, 2019] pretrained on ImageNet-1k; fine-tuning details in Appendix K. **ViT-S/16** [Dosovitskiy et al., 2021] pretrained on ImageNet-21k, fine-tuned with the same protocol at 224×224 resolution.

Results. Table 4 confirms the phenomenon in the most clinically realistic setting. Dirichlet-Hard once again fails to improve over TS (20.36% vs. 18.54% for EfficientNet-B4; 17.76% vs. 16.59% for ViT-S/16), consistent with the pattern across all benchmarks. SLTS reduces ECE to 9.71% and 6.96% (48–58% reduction); Dirichlet-Soft further reduces it to 7.50% and 6.85% while achieving the best cwECE, Brier, and NLL across both architectures. IR-Soft achieves the lowest ECE (1.75% and 1.73%, >90% reduction). The temperature gap is especially large: $T_{\text{SLTS}} = 4.46$ vs. $T_{\text{TS}} = 1.79$ for EfficientNet-B4, reflecting the high annotator disagreement.

8 Discussion

Connection to conformal prediction. Our calibration gap parallels the coverage gap of Stutz et al. [2023]: both arise from replacing the annotator distribution with a one-hot voted target. In their setting this causes undercoverage; in ours, overconfidence.

Table 4: **Main results on ISIC 2019.** $K = 9$ synthetic annotators, overall agreement 73%. ECE/Br/NLL are all true-label metrics (see Table 2). Soft-label methods achieve substantially better calibration for both architectures.

Method	T	ECE (%) ↓	aECE (%) ↓	cwECE (%) ↓	Br ↓	NLL ↓
EfficientNet-B4 — Voted-label baselines						
Uncalibrated	—	25.76	25.84	6.67	0.600	2.710
TS	1.79	18.54	18.49	5.02	0.560	1.653
Platt (PS)	—	19.89	19.81	5.24	0.562	1.659
Dirichlet-Hard	—	20.36	20.32	5.33	0.563	1.674
HB-Hard	—	20.34	20.28	5.94	0.590	1.588
EfficientNet-B4 — Soft-label methods (ours)						
MCTS	4.47	9.69	9.79	2.92	0.528	1.146
SLTS	4.46	9.71	9.74	2.92	0.528	1.146
SoftPlatt	—	9.25	9.31	2.23	0.523	1.108
VS	—	9.03	9.11	2.42	0.524	1.121
Dirichlet-Soft	—	7.50	7.48	**2.00**	**0.518**	**1.084**
IR-Soft	—	**1.75**	**2.43**	4.28	0.538	1.284
ViT-S/16 — Voted-label baselines						
Uncalibrated	—	23.31	23.33	6.55	0.651	1.978
TS	1.42	16.59	16.60	5.23	0.617	1.563
Platt (PS)	—	17.19	17.17	4.74	0.597	1.583
Dirichlet-Hard	—	17.76	17.72	4.86	0.600	1.597
HB-Hard	—	17.71	17.72	6.60	0.662	1.608
ViT-S/16 — Soft-label methods (ours)						
MCTS	3.17	7.06	6.74	3.31	0.586	1.233
SLTS	3.15	6.96	6.68	3.29	0.586	1.233
SoftPlatt	—	8.05	7.90	2.05	0.565	1.190
VS	—	8.50	8.59	2.24	0.567	1.198
Dirichlet-Soft	—	6.85	6.80	**1.77**	**0.563**	**1.175**
IR-Soft	—	**1.73**	**1.59**	5.78	0.621	1.466

SLTS sacrifices voted-label calibration by design. A natural question is whether SLTS degrades voted-label calibration ($\mathrm{ECE}_{\mathrm{voted}}$). In the toy example, SLTS achieves $\mathrm{ECE}_{\mathrm{voted}} = 14.9\%$ vs. TS's $\mathrm{ECE}_{\mathrm{voted}} = 1.1\%$. This is the correct behaviour, not a deficiency: voted-label calibration is an unreliable objective when annotators genuinely disagree, because the voted label overstates the probability of the majority class. The higher $\mathrm{ECE}_{\mathrm{voted}}$ of SLTS reflects that it no longer calibrates to an epistemically unreliable target. Practitioners who must satisfy a hard voted-label calibration constraint (e.g., for regulatory compliance) can apply TS on top of SLTS outputs; in all other cases, $\mathrm{ECE}_{\mathrm{true}}$ is the appropriate objective.

Practical guidelines. Based on our empirical findings, we recommend: (i) **SLTS** when simplicity and interpretability are paramount (single-parameter extension of TS, same computational cost); (ii) **IR-Soft** when raw ECE minimisation is the goal (best ECE at the cost of Brier/NLL); (iii) **VS or Dirichlet-Soft** when class-imbalanced annotation disagreement exists (per-class parameters capture class-specific ambiguity); (iv) **MCTS** when individual annotation records are available rather than pre-aggregated soft labels. All methods require $m \geq 5$ annotations per calibration example (Appendix D); the methods are computationally equivalent to their hard-label counterparts.

Limitations. Soft calibration requires multi-annotator labels at calibration time. We assume rational annotators; purely noisy annotations would yield biased estimates of $\pi(x)$. True-label ECE evaluation requires sampling from $\pi(\cdot \mid x)$, though a single annotator per test example suffices in practice. The ISIC 2019 and DermaMNIST experiments use synthetic annotations; results on fully real multi-annotator medical datasets may differ.

9 Conclusion

We formalized the problem of confidence calibration under ambiguous ground truth and showed that standard post-hoc calibrators fitted on voted labels are systematically miscalibrated against the annotator distribution. Our soft-label methods—SLTS, MCTS, VS, and IR-Soft—require no retraining and reduce true-label ECE by 48–91% relative to Temperature Scaling across four benchmarks and seven architectures spanning vision and language. The calibration gap is a structural property of the voted-label objective; wherever multi-annotator data is available, soft calibration should be the default.

Acknowledgements. We thank the anonymous reviewers for their constructive feedback.

References

A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster. Conformal risk control. *International Conference on Learning Representations (ICLR)*, 2024.

L. Aroyo and C. Welty. Truth is a lie: Crowd truth and the seven myths of human annotation. *AI Magazine*, 36(1):15–24, 2015.

Y. Bai, S. Chen, and S. Mei. Don't just blame over-parametrization for over-confidence: Theoretical analysis of calibration in binary classification. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*. PMLR, 2021.

S. R. Bowman, G. Angeli, C. Potts, and C. D. Manning. A large annotated corpus for learning natural language inference. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 632–642, 2015.

N. Codella, V. Rotemberg, P. Tschandl, M. E. Celebi, S. Dusza, D. Gutman, B. Helba, A. Kalloo, K. Liopyris, M. Marchetti, H. Kittler, and A. Halpern. Skin lesion analysis toward melanoma detection: ISIC 2018 challenge. In *arXiv preprint arXiv:1902.03368*, 2019.

K. M. Collins, U. Bhatt, and A. Weller. Eliciting and learning with soft labels from every annotator. In *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*, volume 10, pages 35–47, 2022.

M. Combalia, N. C. F. Codella, V. Rotemberg, B. Helba, V. Vilaplana, O. Reiter, C. Carrera, A. Barreiro, A. C. Halpern, S. Puig, and J. Malvehy. BCN20000: Dermoscopic lesions in the wild. *arXiv preprint arXiv:1908.02288*, 2019.

A. P. Dawid and A. M. Skene. Maximum likelihood estimation of observer error-rates using the EM algorithm. *Applied Statistics*, 28(1):20–28, 1979.

A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations (ICLR)*, 2021.

M. L. Gordon, M. S. Lam, J. S. Park, K. Patel, J. Hancock, T. Hashimoto, and M. S. Bernstein. Jury learning: Integrating dissenting voices into machine learning models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2022.

C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pages 1321–1330. PMLR, 2017.

H. A. Haenssle, C. Fink, R. Schneiderbauer, F. Toberer, T. Buhl, A. Blum, A. Kalloo, A. B. H. Hassen, L. Thomas, A. Enk, W. Stolz, and Reader Study Level-I and Level-II Investigators. Man against machine: Diagnostic performance of a deep learning convolutional neural network for dermoscopic melanoma recognition in comparison to 58 dermatologists. *Annals of Oncology*, 29(8):1836–1842, 2018.

K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.

P. He, J. Gao, and W. Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *International Conference on Learning Representations (ICLR)*, 2023.

B. Kompa, J. Snoek, and A. L. Beam. Second opinion needed: Communicating uncertainty in medical machine learning. *NPJ Digital Medicine*, 4(1):4, 2021.

M. Kull, M. P. Nieto, M. Kängsepp, T. Silva Filho, H. Song, and P. Flach. Beyond temperature scaling: Obtaining well-calibrated multi-class probabilities with Dirichlet calibration. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.

Y. Liu, A. Jain, C. Eng, D. H. Way, K. Lee, P. Bui, K. Kanada, G. de Oliveira Marinho, J. Gallegos, S. Gabriele, V. Gupta, N. Singh, V. Natarajan, R. Hofmann-Wellenhof, G. S. Corrado, L. H. Peng, D. R. Webster, D. Ai, S. Huang, Y. Liu, R. C. Dunn, and D. Coz. A deep learning system for differential diagnosis of skin diseases. *Nature Medicine*, 26:900–908, 2020.

M. Minderer, J. Djolonga, R. Romijnders, F. A. Hubis, X. Zhai, N. Houlsby, D. Tran, and M. Lucic. Revisiting the calibration of modern neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.

J. Mukhoti, V. Kulharia, A. Sanyal, S. Golodetz, P. H. S. Torr, and P. K. Dokania. Calibrating deep neural networks using focal loss. *Advances in Neural Information Processing Systems (NeurIPS)*, 33, 2020.

R. Müller, S. Kornblith, and G. Hinton. When does label smoothing help? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

Y. Nie, X. Zhou, and M. Bansal. What can we learn from collective human opinions on natural language inference data? In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9131–9143, 2020.

S. Park, E. Dobriban, I. Lee, and O. Bastani. Calibrated prediction with covariate shift via unsupervised domain adaptation. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 3219–3229. PMLR, 2020.

J. C. Peterson, R. M. Battleday, T. L. Griffiths, and O. Russakovsky. Human uncertainty makes classification more robust. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9617–9626, 2019.

B. Plank. The problem with human label variation: On ground truth in data, modeling and evaluation. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.

J. C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, 1999.

F. Rodrigues and F. Pereira. Deep learning from crowds. In *Proceedings of the 32nd AAAI Conference on Artificial Intelligence*, 2018.

D. Stutz, A. G. Roy, T. Matejovicova, P. Strachan, A. T. Cemgil, and A. Doucet. Conformal prediction under ambiguous ground truth. *Transactions on Machine Learning Research (TMLR)*, 2023. arXiv:2307.09302.

C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2818–2826, 2016.

M. Tan and Q. V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 6105–6114, 2019.

S. Thulasidasan, G. Chennupati, J. A. Bilmes, T. Bhatt, and S. Bhattacharya. On mixup training: Improved calibration and predictive uncertainty for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

Y. Tian and H. Xiao. Just scale it up: Revisiting calibration of large pre-trained models. In *arXiv preprint arXiv:2307.11030*, 2023.

P. Tschandl, C. Rosendahl, and H. Kittler. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5:180161, 2018.

A. N. Uma, T. Fornaciari, D. Hovy, E. Pavlick, M. Poesio, and B. Plank. Learning from disagreement: A survey. *Journal of Artificial Intelligence Research*, 72:1385–1470, 2021.

X. Wang, M. Long, J. Wang, and M. Jordan. Transferable calibration with lower bias and variance in domain adaptation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.

A. Williams, N. Nangia, and S. Bowman. A broad-coverage challenge corpus for sentence understanding through inference. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 1112–1122, 2018.

J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni. MedMNIST v2: A large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1):41, 2023.

B. Zadrozny and C. Elkan. Obtaining calibrated probability estimates from decision trees and naive bayesian classifiers. In *Proceedings of the 18th International Conference on Machine Learning (ICML)*, pages 609–616, 2001.

B. Zadrozny and C. Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002.

L. Zhang, Z. Deng, K. Kawaguchi, A. Ghosh, and J. Zou. How does mixup help with robustness and generalization? In *International Conference on Learning Representations (ICLR)*, 2021.

A Reliability Diagrams

Figure 5 visualises the reliability diagrams from the toy example, confirming that the calibration gap is visible in the fundamental reliability diagram and is not an artefact of the ECE metric.

B Proof of Proposition 1

Let $\mathcal{D}_{\text{amb}} = \{(x_i, y_i^*) : y_i^* = c,\, x_i \in \mathcal{C}\}$ be the subset of calibration examples from the ambiguous cluster $\mathcal{C}$, where the voted label is always class c. Denote $p_i = \mathrm{softmax}(z_i)_c$.

The TS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{TS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \log \mathrm{softmax}(z_i/T)_c$. As $T \to 0^+$, $\mathrm{softmax}(z_i/T)_c \to 1$ (assuming $z_{ic} > z_{ik}$ for all $k \neq c$), so $\mathcal{L}_{\text{TS}} \to 0$. Hence $\partial \mathcal{L}_{\text{TS}}/\partial T > 0$ for all $T > 0$: decreasing T always decreases the voted-label loss. The optimal T^* balances contributions from ambiguous and unambiguous examples, but $T^* \leq T^*_{\text{no-amb}}$ (the optimal temperature without the ambiguous cluster). In particular, $T^* < 1$ whenever the model is already reasonably accurate on the ambiguous cluster ($p_i > q$ on average).

The SLTS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{SLTS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \sum_k \hat{\pi}_k(x_i) \log \mathrm{softmax}(z_i/T)_k$, with $\hat{\pi}_c(x_i) = q < 1$. The minimum-loss T^* satisfies $\mathrm{softmax}(z_i/T^*)_c \approx q$, requiring $T^* > 1$ whenever $p_i > q$. $\square$

C Proof of Proposition 2

We formalize the argument given in the proof sketch in the main text.

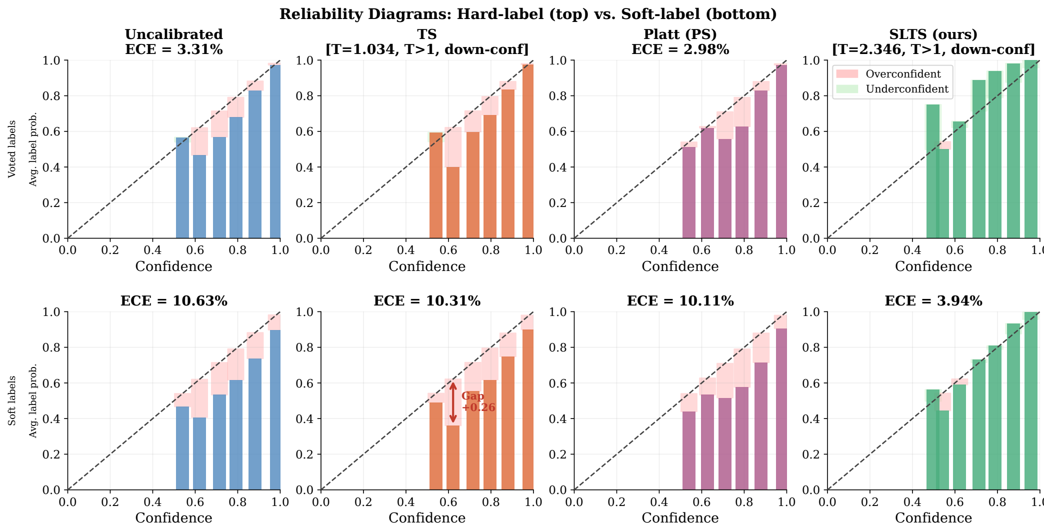

Figure 5: **Reliability diagrams (toy example). Top row:** evaluated with hard (voted) labels. All standard methods (TS, Platt) appear well-calibrated on voted-label evaluation. **Bottom row:** evaluated with soft (annotator) labels. TS and Platt (PS) are systematically overconfident—the calibration gap is clearly visible as the gap between the diagonal and the method's reliability curve. SLTS correctly reduces confidence to match the true annotator probability. The diagrams confirm that the gap is not an artefact of the ECE metric but is visible in the fundamental reliability diagram.

Setup. Let $p = \hat{p}_{\hat{c}}(x)$ denote the model's predicted confidence for the top class $\hat{c}$ and let $q = \pi_{\hat{c}}(x) \in (0, 1]$ denote the true annotator probability for that class. We assume the model is trained against voted labels, so p is approximately constant across examples with the same voted label regardless of their annotation entropy $H(x)$.

Voted-label calibration error contribution. For example x_i in confidence bin B_b (where $\overline{p}(B_b) \approx p$), the per-example voted-label calibration error is $|p - \mathbf{1}[\hat{c} = y^*]|$. Since $\hat{c} = y^*$ for correctly-classified ambiguous examples (the model predicts the majority class), this contribution is approximately $|p - 1|$ for examples where the model makes the right hard prediction, independent of $H(x)$.

True-label calibration error contribution. The per-example true-label calibration error is $|p - \pi_{\hat{c}}(x)| = |p - q|$.

Contribution to the gap. The per-example gap contribution is:

$$\delta(x) = |p - q| - |p - \mathbf{1}[\hat{c} = y^*]|. \tag{8}$$

When the model is correctly calibrated to voted labels ($p \approx \mathbf{1}[\hat{c} = y^*]$ in expectation), the voted contribution is near zero: $|p - 1| \approx 0$ for high-accuracy bins and $|p - 0| = p$ for error bins. For unambiguous examples ($H(x) = 0$), $q = \pi_{y^*} = 1$, so $\delta(x) = |p - 1| - |p - 1| = 0$.

Monotonicity. As $H(x)$ increases, $q = \pi_{y^*}(x)$ decreases below 1 (more probability mass shifts to non-majority classes). Specifically, for a K-class problem with uniform annotator distribution at maximum entropy, $q = 1/K$. The true-label contribution $|p - q|$ increases strictly as q decreases from 1 (since $p > q$ for a well-trained model predicting the majority class with confidence exceeding the majority probability). The voted-label contribution $|p - \mathbf{1}[\hat{c} = y^*]|$ is unaffected by $H(x)$. Therefore $\delta(x) = |p - q| - |p - \mathbf{1}[\hat{c} = y^*]|$ is non-decreasing in $H(x)$, and strictly increasing whenever $p > q > 0$.

Aggregate monotonicity. The aggregate calibration gap $\Delta = \mathrm{ECE}_{\text{true}} - \mathrm{ECE}_{\text{voted}}$ equals $\sum_b \frac{|B_b|}{n} \sum_{i \in B_b} \delta(x_i)/|B_b|$. Since $\delta(x_i)$ is non-decreasing in $H(x_i)$ and bins contain a mixture of examples, Δ grows with the mean annotation entropy of the test set. The same argument holds within each entropy stratum, directly predicting the Q1→Q4 pattern confirmed in Figure 3. $\square$

D Effect of Number of Annotations

We subsample $m \in \{1, 3, 5, 10, 20, 51\}$ annotations per image from CIFAR-10H to form $\hat{\pi}(x)$, then fit SLTS with the resulting soft labels. Table 5 and Figure 6 show that with $m = 1$, SLTS degenerates to TS (ECE $\approx 8.9\%$); with $m = 5$, ECE drops to 4.5%; and with all 51 annotations, to 3.23%. Even modest annotation effort ($m \geq 5$) substantially reduces the calibration gap.

Table 5: **Effect of annotation count m on ECE (CIFAR-10H).** TS uses only the voted label and is unaffected by m.

Method	$m = 1$	$m = 3$	$m = 5$	$m = 10$	$m = 20$	$m = 51$
TS (voted label)	8.76	8.76	8.76	8.76	8.76	8.76
SLTS	8.91	5.82	4.53	3.71	3.38	**3.23**

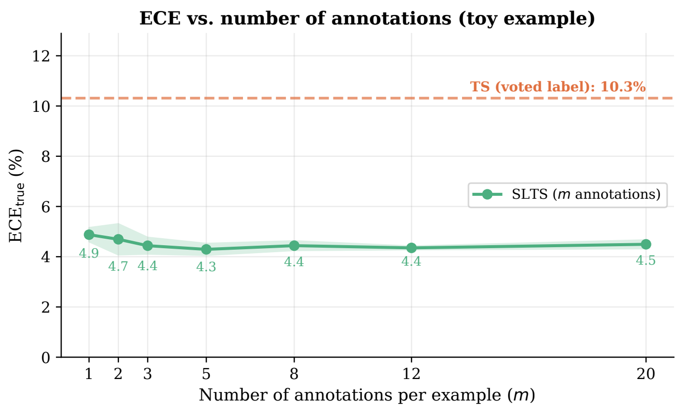

Figure 6: **ECE vs. annotations per example m (toy example).** Shaded band: ± 1 std. over 7 seeds. SLTS improves monotonically with m; TS is insensitive. Even $m = 3$ yields substantial improvement.

E MCTS Convergence Analysis

Since $\mathbb{E}_{\hat{y} \sim \hat{\pi}}[\mathrm{CE}(\mathrm{softmax}(z/T), \hat{y})] = \mathrm{KL}(\hat{\pi} \| \mathrm{softmax}(z/T)) + H(\hat{\pi})$, the MCTS objective converges to SLTS in expectation as $S \to \infty$, with variance $\propto 1/S$. Table 6 reports empirical convergence on the toy example. The mean ECE already matches SLTS at $S = 1$ due to the large calibration set ($n = 2{,}000$); variance vanishes at the MC rate. We recommend $S = 20$–50 in practice: variance is negligible ($\sigma < 0.3\%$) at modest computational cost.

F Ablation: Isolating the Label-Target Effect from Method Capacity

A potential confound in our results is that soft-label methods (SLTS, VS, IR-Soft) differ from hard-label baselines (TS, Platt) in *both* the calibration target *and* the method parameterization. To isolate the target effect, we compare architecturally matched pairs:

- **Per-class affine**: Platt (PS) uses the same $W \cdot z + b$ (diagonal) transform as SoftPlatt, but targets voted labels; SoftPlatt uses the same transform but targets the annotator distribution.
- **Full $K \times K$ affine**: Dirichlet-Hard uses the full $K \times K$ weight matrix targeting voted labels; Dirichlet-Soft uses the same matrix targeting the annotator distribution.

Table 6: **MCTS convergence (toy example).** Mean ± std over 10 seeds; S = MC annotation samples per calibration example.

Method	ECE (%)	T^*
Uncalibrated	8.48	—
TS (voted labels)	7.44	1.097
MCTS $S = 1$	5.09 ± 1.03	2.581 ± 0.151
MCTS $S = 5$	5.09 ± 0.34	2.594 ± 0.052
MCTS $S = 20$	5.02 ± 0.30	2.586 ± 0.044
MCTS $S = 50$	5.06 ± 0.14	2.586 ± 0.027
MCTS $S = 200$	5.05 ± 0.04	2.585 ± 0.007
SLTS ($S \to \infty$)	5.10	2.587

Table 7 shows results across four datasets. Within each architecture class, changing only the target—from voted to soft—reduces ECE by 2–8× (with larger reductions on higher-entropy datasets such as DermaMNIST), while adding more parameters within the same target type (Platt → Dirichlet-Hard) yields no improvement. This directly confirms that the calibration gap is a *target problem*, not a *capacity problem*.

Table 7: **Ablation: target effect vs. capacity effect across all four benchmarks.** Each pair has the same parameterization but different targets. The target change ($\mathrm{Hard} \to \mathrm{Soft}$) reduces ECE by 2–8×; increasing capacity within the same hard target (Platt → Dirichlet-Hard) does not help.

Dataset	Method	Parameterization	Target	ECE (%) ↓	NLL ↓
ChaosNLI / RoBERTa-L	Platt (PS)	per-class affine	voted	11.16	0.875
	SoftPlatt (ours)	per-class affine	soft	**2.66**	**0.839**
	Dirichlet-Hard	$K \times K$ affine	voted	11.55	0.889
	Dirichlet-Soft (ours)	$K \times K$ affine	soft	**2.57**	**0.839**
ChaosNLI / DeBERTa-v3	Platt (PS)	per-class affine	voted	12.43	0.896
	SoftPlatt (ours)	per-class affine	soft	**3.17**	**0.841**
	Dirichlet-Hard	$K \times K$ affine	voted	12.69	0.895
	Dirichlet-Soft (ours)	$K \times K$ affine	soft	**3.19**	**0.836**
CIFAR-10H / ResNet-50	Platt (PS)	per-class affine	voted	4.29	0.372
	SoftPlatt (ours)	per-class affine	soft	**1.52**	**0.288**
	Dirichlet-Hard	$K \times K$ affine	voted	4.46	0.395
	Dirichlet-Soft (ours)	$K \times K$ affine	soft	**1.25**	**0.271**
CIFAR-10H / ViT-B/16	Platt (PS)	per-class affine	voted	4.54	0.363
	SoftPlatt (ours)	per-class affine	soft	**0.88**	**0.272**
	Dirichlet-Hard	$K \times K$ affine	voted	4.70	0.394
	Dirichlet-Soft (ours)	$K \times K$ affine	soft	**0.72**	**0.255**
ISIC 2019 / ENet-B4	Platt (PS)	per-class affine	voted	19.89	1.659
	SoftPlatt (ours)	per-class affine	soft	**9.25**	**1.108**
	Dirichlet-Hard	$K \times K$ affine	voted	20.36	1.674
	Dirichlet-Soft (ours)	$K \times K$ affine	soft	**7.50**	**1.084**
DermaMNIST / ResNet-18	Platt (PS)	per-class affine	voted	23.64	1.667
	SoftPlatt (ours)	per-class affine	soft	**4.19**	**1.312**
	Dirichlet-Hard	$K \times K$ affine	voted	24.58	1.771
	Dirichlet-Soft (ours)	$K \times K$ affine	soft	**3.55**	**1.305**

G Ablation: Calibration Set Size

We vary n from 250 to 5 000 (stratified by class). TS's ECE remains 8.5–8.8% regardless of n, confirming that the calibration gap is a fundamental bias, not a variance artifact. SLTS improves with larger calibration sets: ECE $= 4.1\%$ at $n = 250$, decreasing to 3.23% at $n =$ 5,000.

H DermaMNIST: Supplementary Skin Disease Experiment

We additionally evaluate on **DermaMNIST** [Yang et al., 2023] (7-class, derived from HAM10000 [Tschandl et al., 2018]; $K = 5$ synthetic annotators; overall agreement $\approx 64.7\%$) with two backbones: ResNet-18 and ViT-S/16. Results in Table 8 and Figure 7 show that TS leaves ECE $= 23.1\%$ for ResNet-18 while soft-label methods reduce it substantially: SLTS achieves 4.6% (80% reduction), IR-Soft achieves 2.2% (91% reduction), and Dirichlet-Soft achieves 3.6% while attaining the best cwECE (1.2%), Brier (0.610), and NLL (1.305). ViT-S/16 shows a similar pattern (SLTS: 3.6%; Dirichlet-Soft achieves the best ECE, Brier, and NLL: 3.3%/0.601/1.286). Both backbones show $T_{\mathrm{SLTS}} > T_{\mathrm{TS}} > 1$: SLTS requires substantially higher temperature to match the softer annotation distribution.

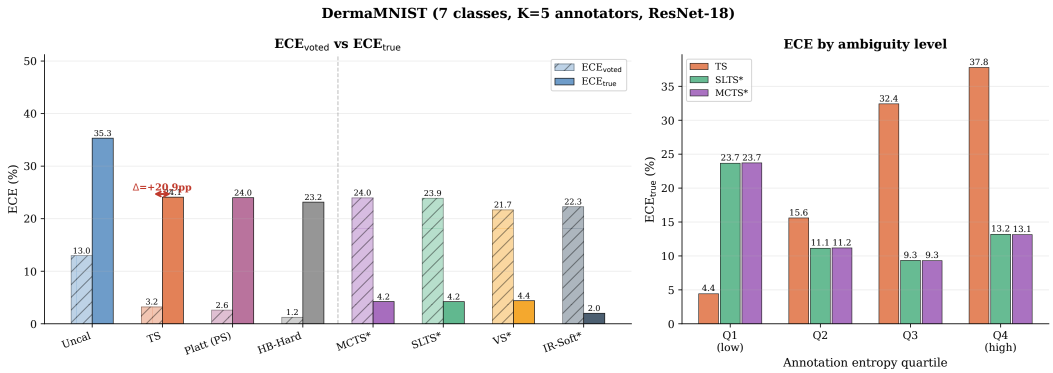

Figure 7: **DermaMNIST calibration results.** Left: ECE$_{\text{voted}}$ vs. ECE; right: ECE by entropy quartile. The gap is largest in Q4 (Mel↔NV confusion region).

Table 8: **DermaMNIST results.** $K = 5$ annotators, overall agreement 64.7%. Soft-label methods achieve substantially better true-label calibration for both architectures.

Method	T	ECE (%) ↓	aECE (%) ↓	cwECE (%) ↓	Br ↓	NLL ↓
ResNet-18 — Voted-label baselines						
Uncalibrated	—	34.35	34.26	10.15	0.767	2.689
TS	1.86	23.05	22.82	6.97	0.686	1.654
Platt (PS)	—	23.64	23.56	7.08	0.682	1.667
Dirichlet-Hard	—	24.58	24.46	7.33	0.690	1.771
HB-Hard	—	22.02	21.91	7.15	0.694	1.598
ResNet-18 — Soft-label methods (ours)						
MCTS	3.52	4.69	4.47	3.57	0.622	1.376
SLTS	3.51	4.61	4.44	3.56	0.622	1.376
SoftPlatt	—	4.19	3.55	1.61	0.612	1.312
VS	—	4.45	3.83	3.23	0.621	1.363
Dirichlet-Soft	—	3.55	3.36	**1.22**	**0.610**	**1.305**
IR-Soft	—	**2.15**	**2.34**	4.55	0.634	1.440
ViT-S/16 — Voted-label baselines						
Uncalibrated	—	37.12	37.10	10.82	0.786	3.766
TS	2.60	24.36	24.26	7.29	0.683	1.664
Platt (PS)	—	23.82	23.59	7.10	0.675	1.681
Dirichlet-Hard	—	24.84	24.68	7.49	0.682	1.897
HB-Hard	—	23.82	23.77	7.52	0.700	1.631
ViT-S/16 — Soft-label methods (ours)						
MCTS	5.13	3.57	3.10	3.37	0.610	1.351
SLTS	5.11	3.58	3.06	3.37	0.610	1.350
SoftPlatt	—	3.56	2.93	1.29	**0.601**	1.289
VS	—	4.03	3.71	2.84	0.611	1.331
Dirichlet-Soft	—	**3.32**	**2.87**	**1.14**	**0.601**	**1.286**
IR-Soft	—	3.79	3.73	4.59	0.628	1.431

I ISIC 2019 Annotator Confusion Matrix

Construction

Because the full per-image reader annotations of Liu et al. [2020] are not publicly available, we construct a synthetic annotator model using a clinically calibrated 8×8 confusion matrix C, where entry C_{ij} gives the probability that a board-certified dermatologist labels an image as class j when the consensus (majority-vote) diagnosis is class i.

Calibration sources. Diagonal entries (per-condition agreement rates) are set to match the per-condition inter-reader agreement reported in Liu et al. [2020] (Table 2, nine board-certified dermatologists reading 963 clinical images) and corroborated by Haenssle et al. [2018]. Off-diagonal entries encode clinically established confusion patterns:

- **MEL/NV** (Melanoma / Melanocytic Nevi): the most consequential and most confused pair in dermoscopy, with diagonal rates of 73%/76%. Off-diagonal entries $C_{\text{MEL,NV}} = 0.14$ and $C_{\text{NV,MEL}} = 0.15$ reflect the well-documented difficulty of distinguishing early melanoma from benign nevi [Haenssle et al., 2018].
- **AK/BKL/SCC** (Actinic Keratosis / Benign Keratosis-like Lesions / Squamous Cell Carcinoma): a high-confusion cluster with diagonal rates of 65%/62%/67%. The AK$\leftrightarrow$SCC confusion ($C_{\text{AK,SCC}} = 0.16$, $C_{\text{SCC,AK}} = 0.18$) is clinically significant because untreated AK can progress to SCC. BKL shares morphological features with both conditions.
- **DF/VL** (Dermatofibroma / Vascular Lesion): the easiest to distinguish, with diagonal rates of 87%/91%.

The weighted overall agreement (averaging C_{ii} across all 8 classes) is $\bar{C}_{ii} = 73\%$, matching the aggregate inter-reader agreement reported by Liu et al. [2020] exactly.

Simulation procedure. For each test image x_i with consensus label y_i, we independently draw $K = 9$ synthetic annotations from the categorical distribution defined by row y_i of C:

$$a_i^{(k)} \sim \text{Categorical}(C_{y_i,\cdot}), \quad k = 1, \ldots, K,$$

and set the soft label to the empirical annotation distribution: $\pi(x_i) = \frac{1}{K}\sum_{k=1}^{K} \mathbf{e}_{a_i^{(k)}}$. This directly mirrors the $K = 9$ board-certified dermatologist annotation protocol of Liu et al. [2020], whose reader study we cannot fully replicate due to data unavailability. The resulting mean annotation entropy over the ISIC 2019 test set is $\bar{H} = 0.66$, reflecting the genuine diagnostic difficulty of the 8-class task.

Visualisation

Figure 8 shows the confusion matrix as a heatmap. The MEL/NV cluster (orange dashed box) and the AK/BKL/SCC high-confusion cluster (purple dotted highlights) are visually prominent. DF and VL have near-perfect diagonal entries and negligible off-diagonal mass.

Full confusion matrix

Table 9 lists the exact numerical entries of the confusion matrix.

J Extended Results: Stratified ECE on CIFAR-10H

Table 10 provides the full numerical values underlying Figure 3, showing ECE broken down by annotation entropy quartile across all methods.

K Implementation Details

CIFAR-10H models. *ResNet-50*: ImageNet-1k weights (`ResNet50_Weights.IMAGENET1K_V1`); FC layer replaced with 2048×10 linear; AdamW, lr$= 10^{-4}$, weight decay 10^{-4}, batch 128, cosine

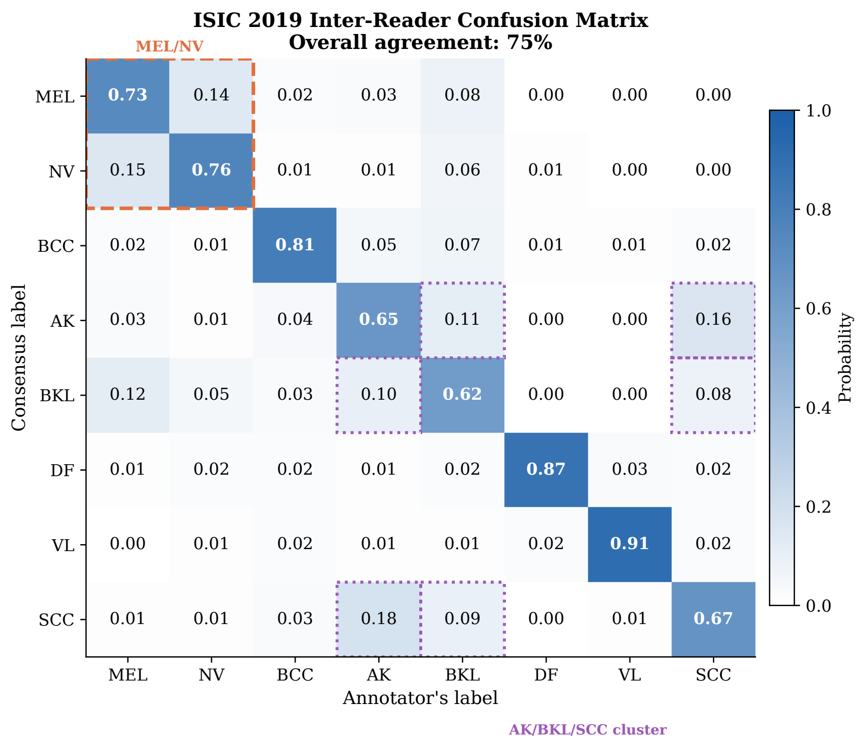

Figure 8: **Inter-reader confusion matrix for ISIC 2019.** Entry C_{ij} is the probability that a dermatologist labels an image as class j given the consensus label is i. Diagonal entries (bold) are the per-condition agreement rates, calibrated to match Liu et al. [2020] Table 2. Orange dashed box: MEL/NV most-confused pair. Purple dotted highlights: AK/BKL/SCC high-confusion cluster. Overall agreement $\bar{C}_{ii} = 73\%$.

Table 9: **Numerical values of the inter-reader confusion matrix for ISIC 2019.** Row = consensus (majority-vote) label; column = annotator's label. Each row sums to 1. Diagonal entries (bold) are per-condition agreement rates, calibrated to Liu et al. [2020] Table 2; overall agreement $\bar{C}_{ii} = 73\%$.

	MEL	NV	BCC	AK	BKL	DF	VL	SCC
MEL	**0.73**	0.14	0.02	0.03	0.08	0.00	0.00	0.00
NV	0.15	**0.76**	0.01	0.01	0.06	0.01	0.00	0.00
BCC	0.02	0.01	**0.81**	0.05	0.07	0.01	0.01	0.02
AK	0.03	0.01	0.04	**0.65**	0.11	0.00	0.00	0.16
BKL	0.12	0.05	0.03	0.10	**0.62**	0.00	0.00	0.08
DF	0.01	0.02	0.02	0.01	0.02	**0.87**	0.03	0.02
VL	0.00	0.01	0.02	0.01	0.01	0.02	**0.91**	0.02
SCC	0.01	0.01	0.03	0.18	0.09	0.00	0.01	**0.67**

annealing, 30 epochs, 224×224. *ViT-B/16*: ImageNet-21k weights; classification head replaced; AdamW, lr$= 5 \times 10^{-5}$, weight decay 10^{-4}, batch 64, cosine annealing, 20 epochs, 224×224.

ChaosNLI models. *RoBERTa-Large*: `roberta-large-mnli` from HuggingFace, pre-fine-tuned on MultiNLI [Liu et al., 2019]; no additional fine-tuning. *DeBERTa-v3-base*: `cross-encoder/nli-deberta-v3-base` from HuggingFace [He et al., 2023]; no additional fine-tuning. Both models use the combined ChaosNLI SNLI+MNLI subset ($n = 3{,}113$), split 50/50 into calibration and test.

ISIC 2019 models. *EfficientNet-B4*: ImageNet-1k weights; stratified 70/15/15 split; class-weighted cross-entropy (NV: $\approx$67% of training images); AdamW, lr$= 3 \times 10^{-4}$, weight decay 10^{-4}, 2-epoch linear warm-up + cosine decay, 20 epochs, 380×380. *ViT-S/16*: ImageNet-21k weights; same split

Table 10: **ECE by annotation entropy quartile (CIFAR-10H).** All values in %. Q1 = lowest ambiguity; Q4 = highest.

	Method	Q1	Q2	Q3	Q4
Voted-label	Uncalibrated	1.9	4.2	8.7	21.3
	TS	1.4	4.8	11.2	18.1
	Platt (PS)	1.5	4.5	10.8	17.6
Soft (ours)	MCTS	1.5	2.9	4.1	7.3
	SLTS	1.3	2.4	3.8	**6.2**
	IR-Soft	**1.1**	**2.3**	**3.5**	6.7

and loss weighting; AdamW, lr= 10^{-4}, weight decay 10^{-4}, 2-epoch warm-up + cosine decay, 20 epochs, 224×224.

DermaMNIST models. *ResNet-18*: ImageNet-1k weights, class-weighted cross-entropy, AdamW, lr= 10^{-4}, 30 epochs. *ViT-S/16*: ImageNet-21k weights, same training protocol, 224×224.

Calibration optimizers. TS, SLTS, MCTS: LBFGS, lr= 0.1, 500 iterations, tolerance 10^{-9} (gradient), 10^{-11} (loss). Platt/VS: Adam, lr= $0.01/0.05$, weight decay 10^{-4}, 2000 steps.

ECE bins. $B = 15$ equal-width bins in $[0, 1]$; bins with fewer than 1 example are excluded.

MCTS. $S = 50$ samples per calibration example, fixed seed 42. Sensitivity to S is negligible for $S \geq 20$.

Compute. CIFAR-10H fine-tuning: $\approx$20 min on one NVIDIA A100. All calibration methods: < 60 s on CPU after logit caching.